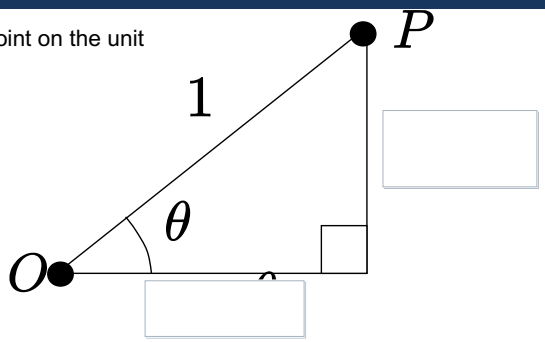
1.42 Trigonometry: Trigonometric Identities in Degrees

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Trigonometric Identities

12

Returning to our point on the unit circle…



1 **Then $\tan\theta =$**

2 Pythagoras gives you...

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Application of identities #1: Proofs

13

Prove that $1 - \tan\theta \sin\theta \cos\theta \equiv \cos^2\theta$

$LHS =$

$= 1 -$

$= 1 - \sin$

$= \cos^2\theta$

$$\tan x = \frac{\sin(x)}{\cos(x)}$$
$$\sin^2 x + \cos^2 x = 1$$

Recall that $\equiv$ means 'equivalent to', and just means the LHS is **always** equal to the RHS for all values of θ.
From Chapter 7 ('Proofs') we saw that usually the best method is to manipulate one side (e.g. LHS) until we get to the other (RHS).

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More Examples

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Prove that $\tan\theta + \frac{1}{\tan\theta} \equiv \frac{1}{\sin\theta\cos\theta}$

$LHS \equiv$ [ ] $- \frac{1}{\sin\theta\cos\theta} + \frac{\cos^2\theta}{\sin\theta\cos\theta}$

$\equiv$ [ ]

[ ]

[ ]

Simplify $5 - 5\sin^2\theta$

[ ]

Tip #2: In any addition/subtraction involving at least one fraction (with trig functions), always combine algebraically into one.

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Test Your Understanding

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Prove that $\frac{\tan x \cos x}{\sqrt{1-\cos^2 x}} \equiv 1$

[ ]

Prove that $\frac{\cos^4\theta - \sin^4\theta}{\cos^2\theta} \equiv 1 - \tan^2\theta$

L [ ]

$\equiv$

$\equiv$

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Prove that $\tan^2\theta \equiv \frac{1}{\cos^2\theta} - 1$

[ ]

1.42 Trigonometry: Trigonometric Identities in Degrees

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10 In each of the following, eliminate θ to give an equation relating x and y:

a $x = \sin\theta, y = \cos\theta$ **b** $x = \sin\theta, y = 2\cos\theta$

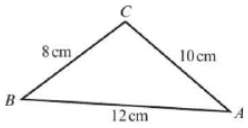
c $x = \sin\theta, y = \cos^2\theta$ **d** $x = \sin\theta, y = \tan\theta$

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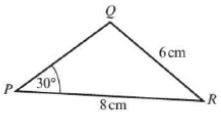
11 The diagram shows the triangle ABC with $AB = 12\,\text{cm}$, $BC = 8\,\text{cm}$ and $AC = 10\,\text{cm}$.



a Show that $\cos B = \frac{9}{16}$. \hfill (3 ma

b Hence find the exact value of $\sin B$.

12 The diagram shows triangle PQR with $PR = 8\,\text{cm}$, $QR = 6\,\text{cm}$ and angle $QPR = 30°$.



a Show that $\sin Q = \frac{2}{3}$.

b Given that Q is obtuse, find the exact value of $\cos Q$.

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Quadratics in sin/cos/tan

29

We saw that an equation can be 'quadratic in' something, e.g. $x - 2\sqrt{x} + 1 = 0$ is 'quadratic in $\sqrt{x}$', meaning that $\sqrt{x}$ could be replaced with another variable, say y, to produce a quadratic equation $y^2 - 2y + 1 = 0$.

Solve $5\sin^2 x + 3\sin x - 2 = 0$ in the interval $0 \leq x \leq 360^\circ$.

Method 1: Use a substitution.

Method 2: Factorise without substitution.

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More Examples

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Solve $\tan^2\theta = 4$ in the interval $0 \leq x \leq 360^\circ$.

Solve $2\cos^2 x + \sin x = 3\sin^2 x$ in the interval $-180^\circ \leq x \leq 180^\circ$.

Tip: We have an identity involving sin^2 and cos^2, so it makes sense to change the squared one that would match all the others.

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Test Your Understanding

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(a) Show that the equation $5\sin x = 1 + 2\cos^2 x$ can be written in the form $2\sin^2 x + 5\sin x - 3 = 0$. (2)
(b) Solve, for $0 \leq x < 360^\circ$, $2\sin^2 x + 5\sin x - 3 = 0$. (4)

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Exercise 10F – Y1 P221-222

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5 Find all the solutions, in the interval $0 \leq x \leq 360^\circ$, to the equation $8\sin^2 x + 6\cos x - 9 = 0$ giving each solution to one decimal place. (6 marks)

6 Find, for $0 \leq x \leq 360^\circ$, all the solutions of $\sin^2 x + 1 = \frac{7}{2}\cos^2 x$ giving each solution to one decimal place.

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Exercise 10F – Y1 P221-222

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7 Show that the equation $2\cos^2 x + \cos x - 6 = 0$ has no solutions.

8 a Show that the equation $\cos^2 x = 2 - \sin x$ can be written as $\sin^2 x - \sin x + 1 = 0$. (2 marks)

b Hence show that the equation $\cos^2 x = 2 - \sin x$ has no solutions. (3 marks)

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Exercise 10F – Y1 P221-222

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9\quad $\tan^2 x - 2\tan x - 4 = 0$.
a Show that $\tan x = p \pm \sqrt{q}$ where p and q are numbers to be found.
b Hence solve the equation $\tan^2 x - 2\tan x - 4 = 0$ in the interval $0 \leq x \leq 540°$.